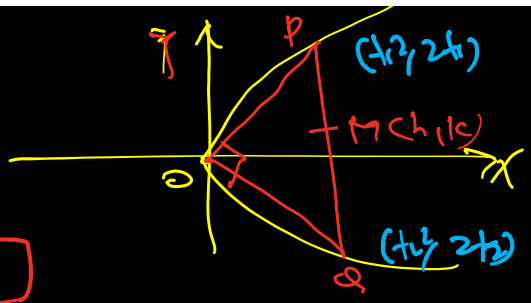
Let O be the vertex of the parabola $y^2 = 4x$ and its chords OP and OQ are perpendicular to each other. If the locus of the mid-point the line segment PQ is a conic C, then the length of its latus rectum is :

A 1 B 2 C 4 D 8

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$\frac{2t_1 - 0}{t_1^2 - 0} \times \frac{2t_2 - 0}{t_2^2 - 0} = -1 \Rightarrow t_1 t_2 = -4$

$t_1^2 + t_2^2 = 2h$ — (I) & $2t_1 + 2t_2 = 2k$ — (II)

$(t_1 + t_2)^2 - 2t_1 t_2 = 2h \Rightarrow k^2 + 8 = 2x$

$y^2 = 2x - 8 \Rightarrow y^2 = 2(x - 4) \Rightarrow y^2 = 4 \times \frac{1}{2}(x - 4)$

$a = \frac{1}{2}$ · L.R. $= 4 \times \frac{1}{2} = 2$ Ans